IA - 4

Take home assignment

Last day for submission: 03/11/2025

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1. A free space-silver interface has $E_0^i = 100$ V/m on the free-space side. The frequency is 15 MHz and the silver constants are $\epsilon_r = \mu_r = 1$ and $\sigma = 61.7$ MS/m. Determine E_0^r and E_0^t at the interface.

2. Find the amplitude of displacement current density in metallic conductor at 60Hz if $\varepsilon = \varepsilon_0$, $\mu = \mu_0$, $\sigma = 5.8 \times 10^7$ S/m and $J = \sin(377\,t - 117.1\,z)\,\mathbf{a}_x$ mA/m^2.

3. In phasor form the electric filed intensity of a uniform plane wave in free space is expressed as $E = (40 - j30)e^{-j20z}\,a_x$ V/m. Find
   a) Angular frequency ω
   b) Phase constant β
   c) Frequency f
   d) Wavelength λ
   e) Magnetic field intensity H

4. A 1 MHz plane wave is propagating in fresh water having $\mu_r = 1$, $\epsilon_r = 81$ and $\epsilon'' = 0$ in $\mathbf{a}_z$ direction. Determine (i) phase constant β, (ii) wavelength, (iii) velocity, (iv) impedance of the medium, and (v) expressions of electric and magnetic fields. Assume the amplitude $E_{x0} = 1$.

5. Given a non-magnetic material having $\varepsilon_r' = 3.2$ and $\sigma = 1.5 \times 10^{-4}$ S/m. Find the values at 3 MHz for (i) loss tangent (ii) attenuation constant (iii) phase constant (iv) Intrinsic impedance.

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